

RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

7. Microwave Antennas and Propagation

3 hrs · asked in **23 of 24 papers** · 3 syllabus sub-topics · **162 marks, 8.4% of 24 papers** → almost every mark is on sub-topic (c), radiation hazards; antenna types and propagation have never been examined.

READ THIS FIRST

Chapter 7 is the **one guaranteed question** of the paper: only 72 Ash skips it in 24 sets. It comes as a 4 to 10 mark theory answer, usually two parts glued together.

- Radiation fields / zones** (11 papers) → §7.1. Three zones, two boundary formulas, one figure. The BTS and microwave-oven versions put numbers in them: worked in the **numerical companion**.
- How microwaves are hazardous; HERP, HERO, HERF** (17 papers) → §7.2. Always the same chain: non-ionizing → dielectric heating → no internal heat sensors → poorly cooled organs.
- SAR** (3 papers, one of them for 10 marks) → §7.3.
- Safety standards** (4 papers) and **safety practices** (7 papers) → §7.4 and §7.5. Name the body, the frequency range and one number each.

Two things worth knowing before the paper starts:

- Nearly every question is “part A + part B” from the list above. Answer each part under its own heading and draw the zone figure whenever “fields” appears.
- Every limit is built on one number:** a whole-body SAR of **4 W/kg** warms the body by about 1°C. Divide by 10 for workers (0.4 W/kg), by 50 for the public (0.08 W/kg). Quote this and the standards section writes itself.

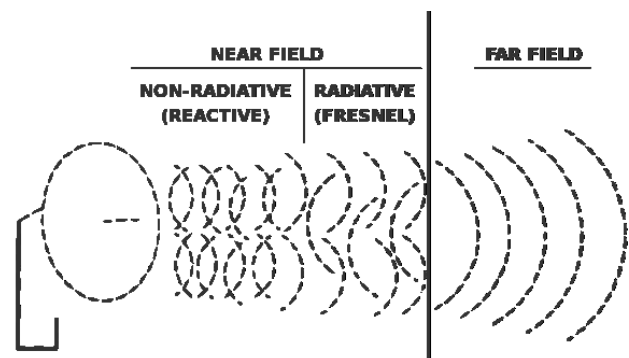
7.1 Microwave Radiation Fields (Radiation Zones)

Name and explain the microwave radiation fields: reactive, radiating near and far field **5 4 3 8 5+5**

TOP 11/24 69 Bh, 71 Ma, 72 Ma, 74 Ma, 76 Bh, 77 Ch, 79 Ch, 80 Ba, 81 Ba, 81 Bh, 82 Ba · oven (74 Ma, 77 Ch) and BTS (81 Ba, 81 Bh, 82 Ba) versions

- Radiation** = energy carried away by EM waves. Round any antenna (or leaking oven, or waveguide opening) the field is split into **three zones** by distance R , antenna largest dimension D and wavelength λ .
- Near field** = the first two zones. **Far field** = the third.

Zone	Boundary
Reactive near field	$R < 0.62\sqrt{D^3/\lambda}$
Radiating near field (Fresnel)	$0.62\sqrt{D^3/\lambda} < R < 2D^2/\lambda$
Far field (Fraunhofer)	$R > 2D^2/\lambda$



Reactive and radiating near field, then far field. Karkee, Ch-7 PDF p5 (white on black in the source, inverted for print).

What each zone is like (the “differentiate” answer, 79 Ch, 72 Ma)

	Reactive near field	Radiating near field	Far field
Energy	stored, sloshes back and forth to the antenna each cycle	radiated, but beam still forming	radiated, travelling away
E and H	90° out of phase, not in fixed ratio	in phase, ratio settling	in phase, $E \perp H \perp$ direction
E/H	not 377Ω ; measure E and H separately	approaching 377Ω	$= 377 \Omega$, so $S = E^2/377$
Fall with R	fast ($1/R^2$ to $1/R^3$)	irregular, local peaks	$S \propto 1/R^2$
Pattern	not defined	changes with distance	fixed shape
Hazard	highest : body close enough couples to stored energy	high: full beam power in a column about D wide	lowest, and predictable from $S = P_t G / 4\pi R^2$

- **Hazardous vs safe** (deck): reactive and radiating near field are treated as **hazardous**, the far field as the **safe** side, where exposure falls as $1/R^2$ and can be checked against a limit (§7.4).
- **Validity** verified/added : the two formulas assume an antenna **larger than a wavelength**. For a small antenna ($D < \lambda$, e.g. a quarter-wave whip) they give a “far field” that starts inside the reactive region. Use instead: reactive field out to $\lambda/2\pi$, far field beyond about 2λ . The 82 Ba quarter-wave question and the deck’s 150 MHz example fall here (ant.py).

Radiation zones of a microwave oven; radiation fields round a GSM BTS 3 4 4 4

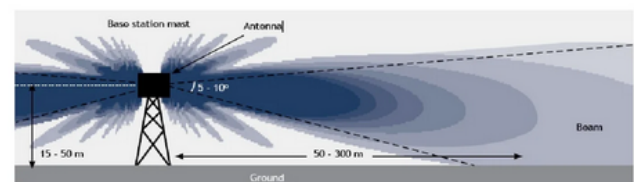
74 Ma, 77 Ch (oven); 81 Ba, 81 Bh, 82 Ba (BTS) · full working in the numerical companion

Microwave oven, $f = 2.45$ GHz, $D = 0.5$ m

- $\lambda = 0.1224$ m, $D > \lambda$ so the formulas hold.
- Reactive: $R < 0.63$ m. Fresnel: 0.63 to 4.08 m. Far field: $R > 4.08$ m.
- **In practice** the cavity is a closed metal box (Faraday cage) and the door mesh holes are far smaller than $\lambda = 12$ cm, so almost nothing leaks. Leakage limit (US FDA): **1 mW/cm^2 at 5 cm** when sold, 5 mW/cm^2 over its life. verified/added
- Zones matter only for leakage at a damaged door seal or hinge: stay out of the reactive zone (arm’s length) while it runs.

GSM / LTE base station (BTS)

- Panel antenna on a mast, main beam tilted a few degrees down, hitting the ground tens to hundreds of metres away (figure).
- **Reactive and Fresnel zones**: within a few metres **in front of** the panel, reached only by riggers on the tower or a rooftop → **occupational** exclusion zone.
- **Far field**: everywhere the public is. Power density there is far below the limit (81 Bh: zones 0.65 m and 4.33 m).
- **Radiation parameters** (81 Ba): frequency, transmit power, antenna gain, EIRP, beam tilt and beamwidth, polarization, power density (W/m^2), field strength (V/m), SAR.



Propagation of “main beam” from antenna mounted on a tower or roof top

Main beam of a mast-mounted antenna. KEC Ch-7 deck, slide 32.

7.2 How Microwave Radiation Is Hazardous: HERP, HERO, HERF

How are microwaves hazardous to humans (living body)? Microwave radiation hazards (short note)

3 4 5 6 8

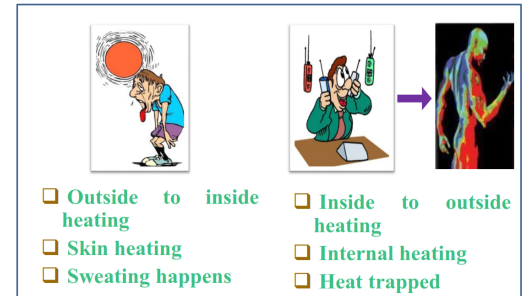
TOP 17/24 69 Bh, 70 Bh, 70 Ma, 71 Bh, 72 Ma, 73 Ma, 73 Bh, 74 Ma, 74 Bh, 75 Bh, 77 Ch, 78 Ch, 79 Bh, 80 Bh, 81 Bh, 82 Ba, 82 Bh · **the most repeated ask of the chapter**

1. Microwaves are non-ionizing

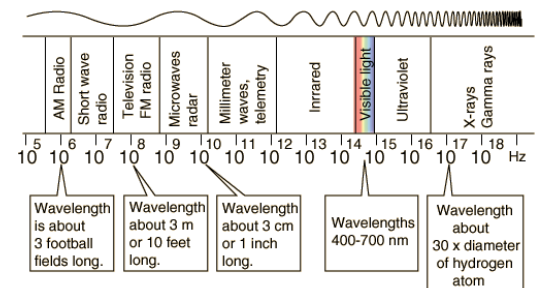
- Photon energy $E = hf$: at 2.45 GHz only 1.0×10^{-5} eV.
- Ionizing a molecule needs about 10 eV → a microwave photon is a **million times** too weak. It cannot break bonds or DNA directly (unlike X-rays, gamma rays).
- So the established hazard is **heat**, not ionization.

2. Thermal effect: dielectric heating

- The field **rotates polar water molecules** and drives **ionic currents** in tissue → friction and I^2R loss → heat.
- Absorbed power per kg = $\sigma E^2 / \rho = \text{SAR}$ (§7.3).
- Microwaves heat **inside out**. Sunlight heats **outside in**, where skin sensors warn and sweat cools (figure).



Sun heating (outside in, felt) vs microwave heating (inside out, trapped). Gangaju, Ch-7 deck, slide 4.



Microwaves sit far below visible light and the ionizing bands. KEC Ch-7 deck, slide 5.

3. Why the heating is dangerous

- No internal heat sensation:** the body senses heat only in the skin, so deep tissue can be damaged **before the person notices**.
- Poorly cooled organs** (little blood flow to carry heat away):
 - Eye lens** → cataract.
 - Testes** → temporary sterility.
 - Stomach, intestines, bladder, gall bladder → thermal damage at high power.
- Body resonance:** absorption peaks when body size is comparable to λ (whole adult body about 70 to 100 MHz). The deck adds: significant absorption even when body size is $\lambda/10$. That is why every limit chart dips lowest at 30 to 300 MHz (§7.4).
- Penetration falls with frequency:** deep at UHF (several cm), about 1 to 2 cm at 2.45 GHz, only the skin above about 10 GHz → lower microwave bands heat deep tissue unfelt, higher bands burn the surface. verified/added
- Long low-frequency magnetic exposure can also induce damaging currents (deck).

4. Other effects

- **RF burns and shocks:** touching a metal object (fence, mast, vehicle) in a strong field drives **contact current** through the hand.
- **Implants:** pacemakers and other active implants can malfunction; metal implants concentrate the field and heat.
- **Microwave hearing:** pulsed radar is heard as clicks (thermoelastic expansion in the head).
- **Behavioural complaints** reported by RF workers: headache, irritability, fatigue, sleeplessness (deck). Not reproduced in blind studies.
- **Cancer:** IARC (WHO) 2011 classes RF fields **Group 2B**, “possibly carcinogenic”, meaning limited evidence. **Correction:** one student deck calls microwaves “known” to cause cancer and another slide calls non-thermal effects “several times more harmful” than thermal; neither is established. verified/added

Link to the zones (69 Bh, 72 Ma, 81 Bh): “how is non-radiating radiation hazardous?” 4 5 8

- The **reactive near field** does not radiate, but its stored E and H fields are strongest there. A body inside it **loads** the antenna and **draws energy out** of the stored field → absorbed as heat, although nothing is “radiated”.
- Radiating near field: full beam power confined to a column about the antenna’s size → highest power density a person in front of the antenna meets.
- Far field: $S = P_t G / 4\pi R^2$ falls with distance → distance is the simplest protection (§7.5).

Types of electromagnetic radiation hazard (RADHAZ); HERP short note 4

74 Ma (types), 80 Bh (HERP)

HERP: to personnel

- Harmful **biological** effects in humans.
- Cause: thermal effect of absorbed energy (points 2 and 3 above).
- Eg: eye lens exposed → too little blood flow to cool it → cataract.
- Controlled by exposure limits (SAR, power density).
- “Hazards of Electromagnetic Radiation to Personnel”.

HERO: to ordnance

- RF induces current in the leads of **electro-explosive devices** (EEDs: detonators, squibs, airbag igniters, missile motors) → **premature firing**.
- Ordnance is **more sensitive** than people: no blood flow to shed heat.
- But easier to protect: metal enclosures reflect the field.
- A chain explosion is possible → HERO limits are **lower** than personnel limits.

HERF: to fuel

- RF-induced **arcs** ignite **fuel vapour** during refuelling near radar or radio transmitters.
- Ignition likely for arcs above about **50 VA**.
- Rules (deck):
 - no transmitting from a vehicle or aircraft being fuelled, or the one beside it
 - no connecting or breaking cables or ground wires while fuelling
 - radar lighting the area at 5 W/cm² peak: shut off
 - antennas of 250 W or less at least 50 ft from fuelling (500 W at sea)

7.3 Specific Absorption Rate (SAR)

Define SAR. Discuss microwave radiation hazards based on SAR exposures; what is non-ionizing SAR

4 6 10

HOT 3/24 74 Bh, 80 Ch, 81 Ba · 80 Ch sets the whole 10 marks on it

Definition

- **SAR** = rate at which RF energy is **absorbed per unit mass** of body tissue, in **W/kg**.

$$\text{SAR} = \frac{d}{dt} \left(\frac{dW}{dm} \right) = \frac{\sigma E^2}{\rho} = c \frac{dT}{dt}$$

- σ = tissue conductivity (S/m), E = **rms field inside the tissue** (V/m), ρ = mass density (**kg/m³**), c = specific heat (J/kg K), dT/dt = initial temperature rise rate.

- **Correction:** the deck prints the density unit as kg/m^2 ; a mass density is kg/m^3 , which is what makes $\sigma E^2 / \rho$ come out in W/kg . verified/added
- **Non-ionizing SAR** (81 Ba): SAR measures **heating** from non-ionizing RF. It is the dose quantity for RF the way absorbed dose (Gy) is for ionizing radiation.

Averaging and measurement

- **Whole-body SAR:** total absorbed power / body mass.
- **Localized (peak spatial) SAR:** averaged over any **1 g** (FCC) or **10 g** (ICNIRP, IEEE 2019) cube of tissue: the phone-against-head case.
- Averaged over time (6 min in ICNIRP 1998).
- **Measured** with a phantom head or body filled with tissue-simulating liquid: a robot moves an E -field probe through it ($\text{SAR} = \sigma E^2 / \rho$), or a temperature probe logs dT/dt ($\text{SAR} = c dT/dt$).
- Eg (ant.py): muscle at 900 MHz, $\sigma \approx 0.94 \text{ S/m}$, $\rho \approx 1040 \text{ kg/m}^3$; $E = 10 \text{ V/m}$ inside gives 0.09 W/kg .

Hazards based on SAR exposure (the 10-mark body)

Whole-body SAR	Effect and what the standards do with it
below 0.08 W/kg	public limit. No established health effect.
0.4 W/kg	occupational limit for trained workers.
about 1 to 4 W/kg	body temperature rises up to about 1°C , thermoregulation (blood flow, sweat) copes in healthy adults.
4 W/kg	threshold: about 1°C core rise, animals show disrupted behaviour. With no heat loss, 4 W/kg warms tissue 1°C in about 15 min ($c \approx 3500 \text{ J/kg K}$).
well above 4 W/kg	heat stress, heat stroke; local hot spots burn tissue, cataract in poorly cooled lens.

- **Safety factors:** threshold $4 \xrightarrow{\div 10} 0.4$ (occupational) $\xrightarrow{\div 5} 0.08 \text{ W/kg}$ (public). The student deck's "one-tenth of the observed threshold SAR (0.4 W/kg)" means the **limit** is 0.4; the threshold is 4. verified/added
- **Localized SAR limits:** public 2 W/kg over 10 g head and trunk (ICNIRP, IEEE 2019), **1.6 W/kg over 1 g** (FCC, US and India phones); workers 10 W/kg.
- **Mobile phones:** SAR is declared by the maker (typically 0.2 to 1.6 W/kg). Using a hands-free kit or speaker cuts it sharply because the head leaves the reactive near field of the handset antenna.

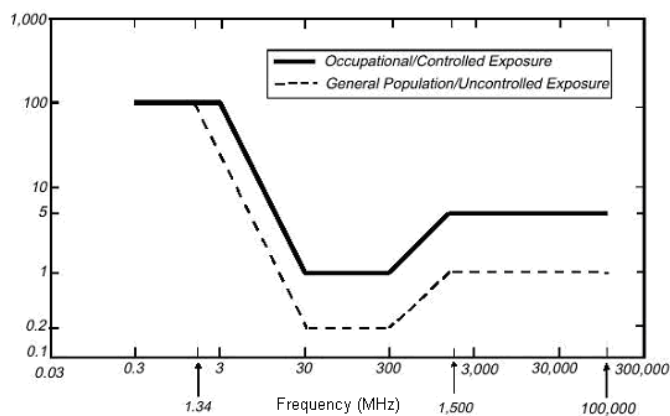
7.4 International EMR Safety Standards

International standards and recommended practices (SARPs) for RF/microwave exposure; IEEE safety standard 2 3 3+3 5+5

HOT 4/24 70 Ma, 76 Bh, 80 Ba, 82 Ba · practices half in §7.5

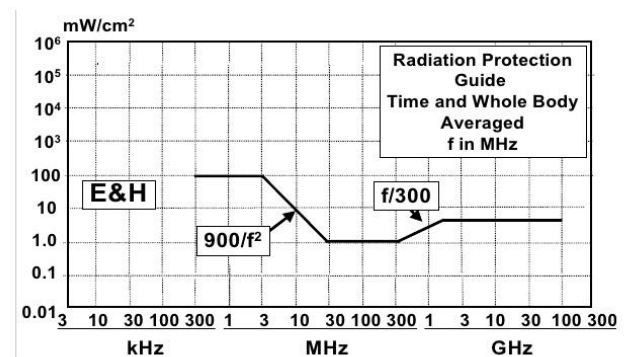
- **Basis of all of them:** adverse effects scale with absorbed power → **basic restriction** in SAR (W/kg), and because SAR cannot be measured in a person, a **reference level** (MPE) in measurable power density (mW/cm^2 , W/m^2) or field strength (V/m , A/m).
- **Two tiers:** **occupational / controlled** (trained workers who know they are exposed) and **general public / uncontrolled** (everyone else, 24 hours a day, including children, the ill and the elderly) → public limit 5 times lower.
- $1 \text{ mW/cm}^2 = 10 \text{ W/m}^2$.

Body and standard	Range	Key numbers
IEEE / ANSI C95.1 (US; 1982, 2005, 2019)	3 kHz to 300 GHz (0 Hz to 300 GHz in 2019)	1982 single tier: 1 mW/cm^2 at 30 to 300 MHz, $f/300$ to 1.5 GHz, 5 mW/cm^2 above. 2005/2019: two tiers, whole body $0.08 / 0.4 \text{ W/kg}$, local $2 / 10 \text{ W/kg}$ over 10 g (2019).
FCC (US, OET Bulletin 65)	300 kHz to 100 GHz	public 0.2 mW/cm^2 at 30 to 300 MHz, $f/1500$ to 1.5 GHz, 1.0 mW/cm^2 above; occupational 5 times higher. Phone SAR 1.6 W/kg over 1 g. Averaging 6 min (workers), 30 min (public).
ICNIRP (international, recognized by WHO; 1998, revised 2020)	100 kHz to 300 GHz	public $f/200 \text{ W/m}^2$ at 400 to 2000 MHz (4.5 at 900, 9 at 1800), 10 W/m^2 above 2 GHz; workers 5 times. SAR $0.08 / 0.4 \text{ W/kg}$ whole body, $2 / 10 \text{ W/kg}$ local. Adopted by most countries.
IRPA / INIRC (1988)		ICNIRP's predecessor: first international occupational and public limits.
OSHA (US workplaces)	10 MHz to 100 GHz	10 mW/cm^2 averaged over 6 min (older guide).
ITU-T K.52, K.70	telecom sites	how to check a BTS site against ICNIRP limits, and how to mitigate (compliance distance, exclusion zones).
WHO EMF Project, IARC		reviews health evidence; IARC Group 2B (2011).

FCC MPE (power density, mW/cm^2 , vs MHz): solid occupational, dashed public.

Karkee PDF p18.

- **Read the charts:** both dip at **30 to 300 MHz**, the body-resonance band (§7.2), and rise again above 300 MHz as absorption becomes shallower.
- **Correction:** the deck's "IRPA permissible exposure level" table (PDF p17: $200 \mu\text{T}$, 5 kV/m , 20 mV/m in the head) is ICNIRP's **50 Hz power-line** limit, not a microwave one. It is left out here. verified/added
- **SARPs (80 Ba) = Standards And Recommended Practices:** answer with the ICNIRP / IEEE limits above as the standards and §7.5 as the practices.



ANSI C95.1-1982 Radiation Protection Guide. Karkee PDF p19.

7.5 RF/Microwave Safety Practices

RF/MW radiation hazards and safety practices; list and describe five major RF safety practices 3 5

6 5+5

TOP 7/24 70 Bh, 71 Bh, 73 Bh, 74 Bh, 76 Bh, 79 Ch, 79 Bh · short-note form mostly; hazard half is §7.2

Five major practices (79 Ch)

1. **Time, distance, shielding:**
 - cut exposure time
 - keep out of the main beam and out of the near field (power density $\propto 1/R^2$)
 - metal panels or mesh, RF absorbers, Faraday enclosures
2. **Engineering controls:**
 - safety **interlocks** that cut RF when a door or panel opens
 - RF gaskets, choke joints, oven door mesh
 - test into a **dummy load**, never an open waveguide
 - grounding straps not bent at sharp angles (they radiate)
3. **Administrative controls:**
 - RF survey of the site, marked **exclusion zones**, warning signs (figure)
 - **lockout / tagout** before work on a tower or transmitter
 - never look into a live waveguide, horn or dish
 - training and exposure records
4. **Personal protective equipment:** RF protective suit (stainless steel woven into fire-retardant fibre) for high-field areas; personal RF monitor that alarms.
5. **Regular inspection:**
 - **visual:** housekeeping, interlocks present, high-voltage grounding rods, condition of cables and coax, RF shielding intact
 - **operational:** RF level survey in occupied spaces, interlock tests, lockout used properly



RF warning sign at an exclusion-zone boundary.

KEC Ch-7 deck, slide 24.

More practices (deck)

- Operate stray-RF sources (induction and dielectric heaters) **remotely**.
- Keep workers below the recommended limits (§7.4).
- **HERO:** keep EEDs away from transmitters.
- **HERF:** no RF devices in flammable atmospheres; fuelling rules of §7.2.
- RF-sensitive equipment and people with **implants** away from sources.
- Switch equipment off when not in use.
- **Public:** antennas high and beams tilted clear of rooftops; hands-free phones; awareness and school teaching.

7.6 Microwave Antennas

Microwave antenna types syllabus, no PYQ yet

syllabus 7(a) · the deck leaves it as an assignment and refers to Pozar

- Conventional wire antennas work at microwaves too, but the short wavelength allows **aperture antennas** a few λ across → high gain, narrow beams in a small size.
- **Parameters:** gain G and directivity, beamwidth, efficiency, polarization, input impedance / VSWR, bandwidth. Aperture gain $G = 4\pi A_e / \lambda^2$ with $A_e = \eta \pi D^2 / 4$.

Type	How it works	Typical use
Horn	flared end of a waveguide; flare matches guide to free space. E-plane, H-plane, pyramidal, conical. 10 to 25 dBi	dish feed, gain standard in measurements
Parabolic reflector	feed (horn) at focus; reflector turns spherical wave into plane wave. $G = \eta(\pi D/\lambda)^2$, HPBW $\approx 70\lambda/D$ degrees. Front-fed or Cassegrain	satellite earth stations, radar, point-to-point links
Lens	dielectric (slows wave) or metal-plate (speeds it) lens collimates like an optical lens	mm-wave radar, automotive
Slot	slot cut in a waveguide wall or ground plane; complement of a dipole (Babinet); flush	aircraft, slotted-guide arrays
Microstrip patch	metal patch about $\lambda_g/2$ long over a ground plane on a substrate; low profile, narrow band (1 to 5%), low power; arrays	phones, GPS, WLAN, 5G arrays
Helical	wire helix; axial mode when circumference $\approx \lambda \rightarrow$ circular polarization	satellite, space telemetry

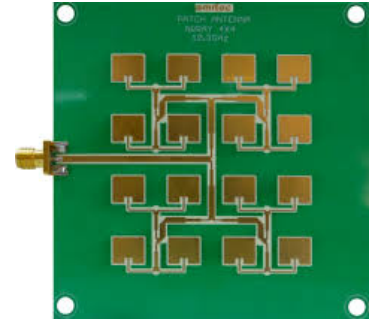
Eg (ant.py): 1.2 m dish at 12 GHz, $\eta = 0.55$: $G = 41.0$ dBi, HPBW = 1.46° .



Pyramidal horn.



Parabolic reflector.



16-element patch array. All three: 2078 student presentation.

- **Corrections to the student deck** verified/added : a microstrip antenna is a **microwave** antenna (at 100 MHz a half-wave patch would be over a metre long), not one “used at low frequency”; and its “MIMO” slide shows a Yagi array. MIMO is a way of using several antennas (spatial multiplexing, diversity), not an antenna type.

Choose a proper microwave measurement tool to test an antenna as a DUT; explain its working principle 8

PIN 1/24 74 Bh · instrument principles in Ch 8

Tool 1: vector network analyzer (VNA), port 1 to the antenna

- Measures S_{11} vs frequency $\rightarrow$ return loss, VSWR = $(1 + |S_{11}|)/(1 - |S_{11}|)$, input impedance on a Smith chart, and the **bandwidth** where VSWR < 2 .
- Working: internal source sweeps; directional couplers split incident and reflected waves; receivers compare amplitude and phase; calibration (short, open, load) removes cable and coupler errors. Detail in Ch 8.

Tool 2: anechoic chamber (pattern and gain)

- Walls lined with RF absorber cones (no reflections). Known source horn at one end, antenna under test on a rotating **positioner** at the other.
- Separation $R > 2D^2/\lambda$ so the test is in the **far field** (§7.1).
- Rotate and record received power (VNA S_{21} or spectrum analyzer) $\rightarrow$ **radiation pattern**, beamwidth, side lobes.
- **Gain**: compare with a standard gain horn, or three-antenna method with Friis.

7.7 Propagation Characteristics of Microwaves

Propagation characteristics: line of sight, atmosphere, ground and plasma effects syllabus, no PYQ yet

syllabus 7(b) · deck refers to Pozar; numbers from `ant.py`

Line-of-sight (space wave)

- Ground wave dies within a short range and the ionosphere does not reflect microwaves (they pass through) → terrestrial links are **LOS**, satellites use the fact that they pass.
- Radio horizon** with $k = 4/3$ earth (refraction bends the ray down): $d \approx 4.12\sqrt{h}$ km, h in m. Two masts: $d = 4.12(\sqrt{h_1} + \sqrt{h_2})$; 30 m and 50 m give 51.7 km.
- Fresnel clearance**: keep 60% of the first Fresnel zone $r_1 = \sqrt{\lambda d_1 d_2 / (d_1 + d_2)}$ clear of obstacles; 50 km hop at 6 GHz needs $r_1 = 25$ m at mid-path.
- Free-space path loss**:
 $L = 32.44 + 20 \log d_{km} + 20 \log f_{MHz}$ dB; Friis
 $P_r = P_t G_t G_r (\lambda / 4\pi d)^2$.

Atmosphere, ground and plasma effects

- Gas absorption**: water vapour peak at 22 GHz, **oxygen at 60 GHz**; little below 10 GHz. The 60 GHz peak suits short private links and satellite crosslinks.
- Rain and fog**: drops comparable to λ scatter and absorb → rain fade, serious above about 10 GHz (Ku, Ka bands).
- Refraction**: ducting and super-refraction extend or bend paths → fading and interference.
- Ground effect**: reflection off ground or water adds a second ray → **multipath fading**; earth bulge blocks long hops.
- Plasma effect**: ionosphere rotates polarization (Faraday rotation) and causes scintillation at low GHz; the plasma sheath round a re-entering spacecraft blocks signals (blackout).
- Diffraction** round obstacles is weak at short λ → shadowing behind buildings and hills.

7.8 PYQ Mapping

Theory questions, covered in this document

- Microwave radiation fields; differentiate reactive, near and far field** → §7.1. 69 Bh 8, 71 Ma 5, 72 Ma 5, 76 Bh 5, 79 Ch 5
- Radiation zones of a microwave oven** → §7.1 + companion Problem 1. 74 Ma 4, 77 Ch 3
- BTS fields: classify, radiation parameters; 900 vs 1800 MHz; hazardous and safe zone at 2600 MHz** → §7.1 + companion Problems 1 and 2. 81 Ba 4, 81 Bh 4, 82 Ba 4
- RF radiation fields to public exposure** → §7.1 and §7.4. 80 Ba 3
- How microwaves are hazardous; microwave radiation hazards (short note)** → §7.2. 69 Bh 8, 70 Ma 5, 72 Ma 5, 73 Ma 5, 75 Bh 5, 77 Ch 3, 78 Ch 5, 81 Bh 4, 82 Ba 2, 82 Bh 4
- Hazards + safety practices (short note)** → §7.2 and §7.5. 70 Bh 5, 71 Bh 5, 73 Bh 6, 74 Bh 6, 79 Bh 3+3
- Types of EM radiation hazard; HERP** → §7.2. 74 Ma 4, 80 Bh 4
- SAR: define, effect as hazard, non-ionizing SAR** → §7.3. 74 Bh 6, 80 Ch 10, 81 Ba 4
- International standards, SARPs, IEEE standard** → §7.4. 70 Ma, 76 Bh 5, 80 Ba 3, 82 Ba 2
- Five major RF safety practices** → §7.5. 79 Ch 5, 76 Bh
- Antenna as DUT: measurement tool and principle** → §7.6. 74 Bh 8

Numerical content

- Radiation-zone boundaries** (BTS at 2600 MHz, BTS at 900 / 1800 MHz with quarter-wave antennas, microwave oven) → numerical companion, 2 problems. Every number is asserted in `src/ant.py`.

Corrections, and fixes made to the PYQ documents while building this chapter

- Karkee PDF p6: SAR mass density is kg/m^3 , not kg/m^2 .
- Karkee PDF p17: the “IRPA” table is a 50 Hz power-line limit.
- Deck example (150 MHz, $D = 0.5$ m): $D < \lambda$, so its zone boundaries are not valid; small-antenna rule applies.
- Student decks: cancer “known” (IARC says possibly, 2B); non-thermal effects “several times more harmful” (not established); “threshold SAR 0.4” (0.4 is the limit, 4 the threshold); microstrip “used at low frequency”; Yagi pictured as “MIMO”.
- Sorted PYQ documents: the hazard short notes were tagged 71 Ma and 72 Ash, which ask none. They are 70 Bh, 71 Bh and 73 Ma (73 Ma was missing). 70 Ma’s note is [5], not [8]. Bold tags for 69 Bh, 71 Bh and 73 Bh made consistent (Regular papers).

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.